Turning historical aviation accident data into actionable safety intelligence

PROJECT OBJECTIVE | TABLEAU DASHBOARD

Executive summary

The project will develop an interactive Tableau dashboard to examine historical aviation accident records across geography, time, aircraft category, flight purpose, flight phase and damage severity. The objective is to identify concentrations and patterns in recorded incidents, equip stakeholders with a consistent performance view, and highlight areas for further safety investigation.

The questions the analysis will answer

- Where are recorded accidents most concentrated, and how has their volume evolved over time?
- Which flight purposes and operational phases account for the largest shares of recorded accidents?
- How are recorded accidents distributed across aircraft categories and damage-severity classifications?

Core performance indicators

INDICATOR	DEFINITION
Total recorded aviation accidents	Number of accident records within the selected reporting period.
US accidents | count and share	Number and percentage of recorded accidents associated with the United States.
Airplane accidents | count and share	Number and percentage of recorded accidents classified in the airplane category.

Dashboard analytical framework

01 Geographic concentration Top 10 airport locations by recorded accident count; identify locations for deeper review.

02 Historical trend Annual recorded accident volume; surface changes in the observed pattern over time.

03 Flight purpose Top 10 flight purposes by recorded accident count; compare activity classifications.

04 Phase of flight Accident count and share of total by operational phase, including takeoff, cruise and landing.

05 Damage severity Accident count and share of total by reported aircraft damage classification.

Decision value and interpretation

The dashboard will provide a structured basis for prioritizing follow-up analysis, monitoring historical patterns and communicating accident distributions to aviation safety stakeholders. Findings should be treated as descriptive signals rather than evidence of causation or comparative safety risk: accident rates require exposure denominators (such as flight hours or departures), and financial impact requires separate cost data.